# Malcolm Certain

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## Education

**Brown University** *Sep. 2022 - Present*
*Sc.B. in Applied Mathematics-Economics and Urban Studies, 4.0 GPA*

## Research Experience

**Undergraduate Research Assistant** *Annenberg Institute*
*Economics of Education* *Brown University*
*Professor Matthew A. Kraft* *Sep. 2024 - Present*

- Did event-study analysis on the effects of weather-related natural disasters on student performance
- Read and summarized technical econometric literature on difference-in-differences for our research team
- Created data visualizations summarizing the effects of climate change on students using a combination of STATA, Python, R, and ArcGIS

**Undergraduate Research Assistant** *Annenberg Institute*
*Political Economy* *Brown University*
*Professors Jesse Bruhn and Thea How Choon* *Jan. 2025 - Present*

- Cleaned and summarized data on language used in city council meetings in the US using Python
- Researched technical literature on shift-share instruments to study causal impacts of national events on local politics
- Replicated results from a paper with a similar empirical strategy using STATA

**Undergraduate Research Fellow** *Annenberg Institute*
*Public Economics* *Brown University*
*Professor Anna Aizer* *May 2024 - Jul. 2024*

- Cleaned and analyzed large amounts of US Census data in STATA
- Created maps showing timing of policy rollout across the US using Python
- Helped other members of the research team learn STATA and debug code

## Teaching Experience

**Undergraduate Teaching Assistant** *Dept. of Economics*
*ECON 1430: The Economics of Social Policy* *Brown University*
*Professor Emily Oster* *Jan. 2025 - Present*

- Assisted students with writing and debugging STATA code
- Helped students select a research design and empirical strategy for an original research paper

**Undergraduate Teaching Assistant** *Dept. of Economics*
*ECON 1130: Intermediate Microeconomics (Mathematical)* *Brown University*
*Dr. Alex Poterack* *Jan. 2024 - May 2024*

- Taught students topics in microeconomics, including utility, game theory, and general equilibrium
- Graded problem sets

**Undergraduate Teaching Assistant** *Dept. of Mathematics*
*MATH 0090, 0100, 0200: Calculus I-III* *Brown University*
*Dr. Yu-Wen Hsu* *Jan. 2023 - May 2024*

- Taught students topics in calculus, including differentiation, integration, optimization, and vector calculus
- Graded problem sets

## Coursework

**Economics**

- ECON 1130: Intermediate Microeconomics
- ECON 1210: Intermediate Macroeconomics
- ECON 1630: Mathematical Econometrics†
- EDUC 1340: Economics of Education
- ECON 1360: Health Economics‡
- ECON 1870: Game Theory†
- ECON 2400: Applied Econometrics*‡

**Mathematics**

- MATH 0200: Multivariable Calculus
- MATH 0520: Linear Algebra†
- APMA 0350: Ordinary Differential Equations
- APMA 0360: Partial Differential Equations
- MATH 1010: Real Analysis†
- MATH 1210: Probability Theory†
- APMA 1941H: Mathematics and Rational Choice†

* *-graduate level*
† *-proof-based*
‡ *-included research component*

## Awards and Honors

Omicron Delta Epsilon *Spring 2025*